EXPERIMENT NO. 4

Count Equal Pairs Whose Index Product Is Divisible by K

Programming Language: Java

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Aim

To write a Java program that counts the pairs (i, j) with nums[i] == nums[j] and (i * j) divisible by k.

Question

Given a 0-indexed integer array nums of length n and an integer k, return the number of pairs (i, j) where 0 <= i < j < n, such that nums[i] == nums[j] and (i * j) is divisible by k.

Example 1

Input: nums = [3,1,2,2,2,1,3], k = 2

Output: 4

Explanation: The valid pairs are (0,6), (2,3), (2,4) and (3,4): the values are equal and the products 0, 6, 8 and 12 are divisible by 2.

Example 2

Input: nums = [1,2,3,4], k = 1

Output: 0

Explanation: No value is repeated, so there are no pairs that meet the requirements.

Algorithm

Step 1: Start.

Step 2: Read the array nums of size n and the integer k, and set count = 0.

Step 3: For each index i from 0 to n-2, run j from i+1 to n-1.

Step 4: If nums[i] == nums[j] and (i * j) % k == 0, increment count.

Step 5: Print count.

Step 6: Stop.

Program (Java)

```
public class Main {

    static int countPairs(int[] nums, int k) {
        int n = nums.length, count = 0;
        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                if (nums[i] == nums[j] && (i * j) % k == 0)
                    count++;
```

```
            }
        }
        return count;
    }

    public static void main(String[] args) {
        int[] nums1 = {3, 1, 2, 2, 2, 1, 3};
        int[] nums2 = {1, 2, 3, 4};

        System.out.println("Output: " + countPairs(nums1, 2));   // Example 1
        System.out.println("Output: " + countPairs(nums2, 1));   // Example 2
    }
}
```

Output

```
Output: 4
Output: 0

=== Code Execution Successful ===
```

Complexity Analysis

Time Complexity: O(n^2) - every pair (i, j) with i < j is checked once.
Space Complexity: O(1)

Result

Thus the program was executed successfully and the output was verified.